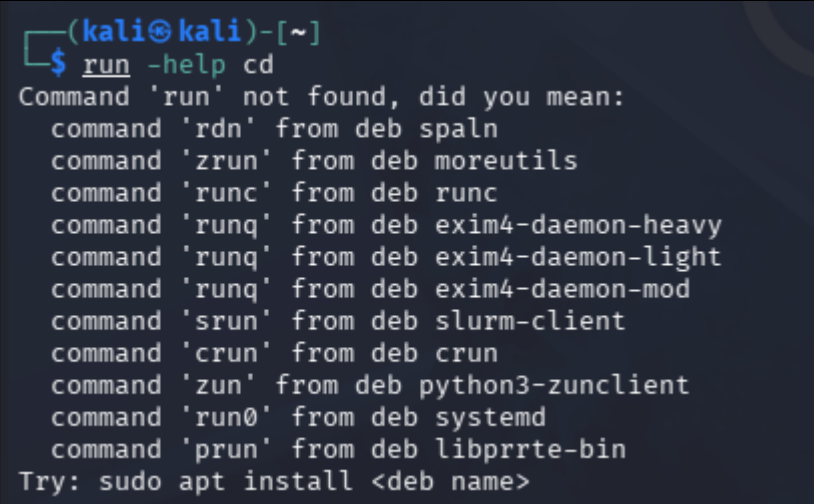


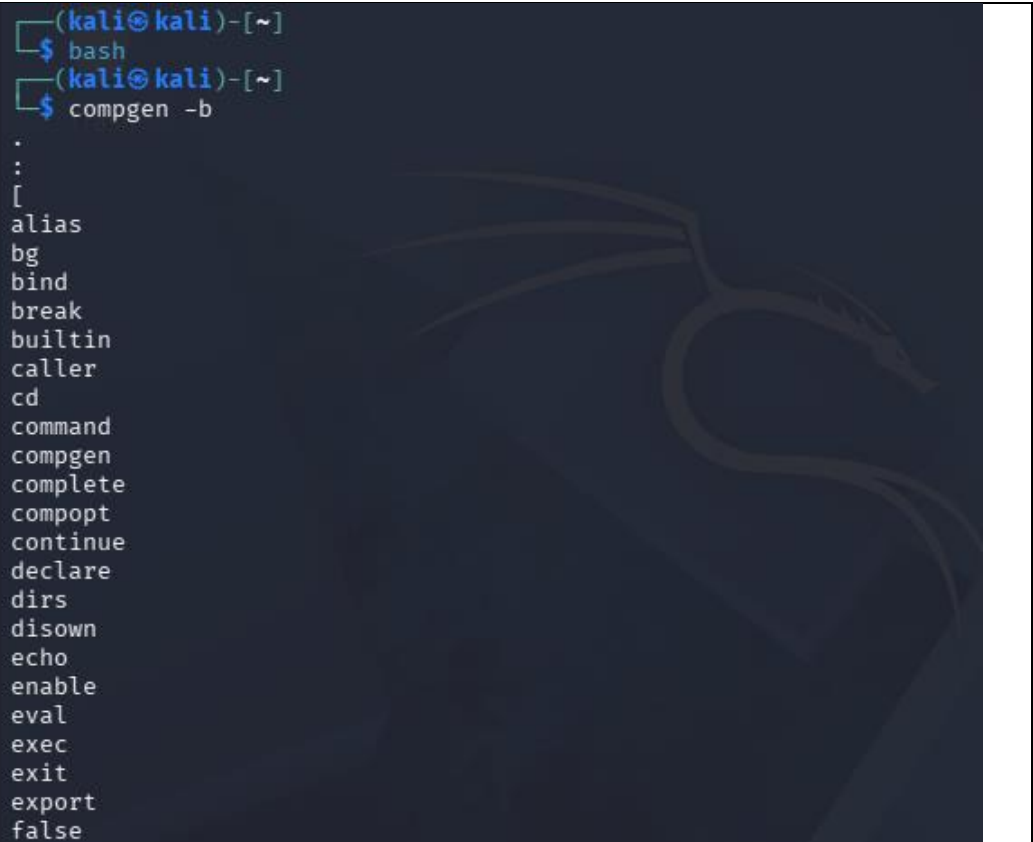
LoS Lab Assignment – 2

1. Getting Help:

Task	Command Name	Syntax	Example	Screenshots
To get manual page for the known command	man	man <command>	Man mkdir	 The screenshot shows a terminal window displaying the man page for the 'mkdir' command. The window title is 'Session Actions Edit View Help'. The content includes sections for NAME, SYNOPSIS, DESCRIPTION, OPTIONS, AUTHOR, REPORTING BUGS, and SEE ALSO. The terminal background is dark with light-colored text. A 'KALI' logo is visible in the background of the terminal window.

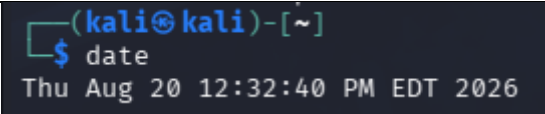
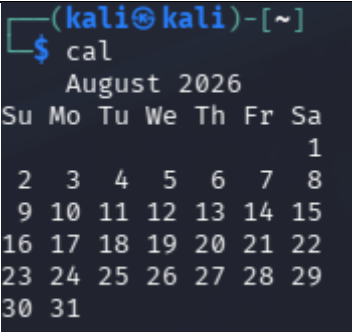
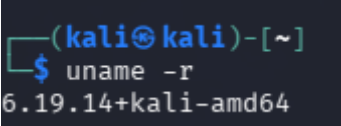
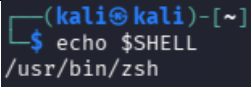
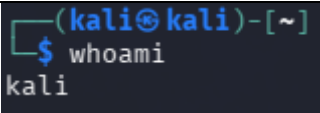
To get manual page for the unknown command	man	Man <command>	Man mklr	<pre>(kali㉿kali)-[~] \$ man mklr No manual entry for mklr (kali㉿kali)-[~] \$</pre>
To know the source file binary	which	which<command>	which ls	<pre>(kali㉿kali)-[~] \$ which ls ls: aliased to ls --color=auto (kali㉿kali)-[~] \$</pre>
To know the path of the command	whereis	whereis <command>	whereis bash	<pre>(kali㉿kali)-[~] \$ whereis bash bash: /usr/bin/bash /usr/share/man/man1/bash.1.gz (kali㉿kali)-[~] \$</pre>
To know the command is external or internal	type	type <command>	type ls	<pre>(kali㉿kali)-[~] \$ type ls ls is an alias for ls --color=auto (kali㉿kali)-[~] \$</pre>

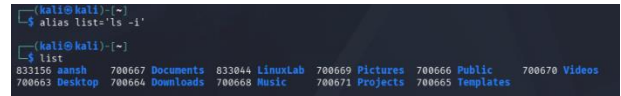
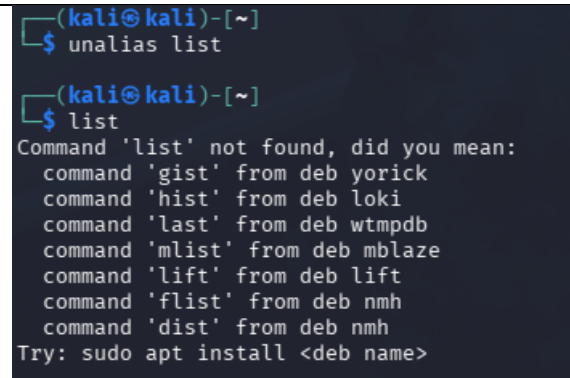
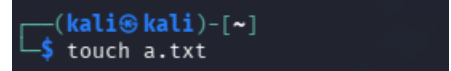
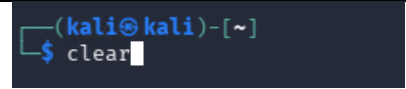
To get help for the internal command	help	Run help-command	Run-help cd	 <pre>(kali㉿kali)-[~] \$ run -help cd Command 'run' not found, did you mean: command 'rdn' from deb spaln command 'zrun' from deb moreutils command 'runc' from deb runc command 'runq' from deb exim4-daemon-heavy command 'runq' from deb exim4-daemon-light command 'runq' from deb exim4-daemon-mod command 'srun' from deb slurm-client command 'crun' from deb crun command 'zun' from deb python3-zunclient command 'run0' from deb systemd command 'prun' from deb libprte-bin Try: sudo apt install <deb name></pre>
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To list out bash commands	compgen	bash compgen -b	Bash compgen -b	 <pre>(kali㉿kali)-[~] \$ bash (kali㉿kali)-[~] \$ compgen -b . : [alias bg bind break builtin caller cd command compgen complete compt continue declare dirs disown echo enable eval exec exit export false</pre>
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To know the usage of the command	--help	command --help	cp --help	<pre> (kali@kali)-[~] \$ cp --help Usage: cp [OPTION]... [-T] SOURCE DEST or: cp [OPTION]... SOURCE... DIRECTORY or: cp [OPTION]... -t DIRECTORY SOURCE... Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY. Mandatory arguments to long options are mandatory for short options too. -a, --archive same as -dR --preserve=all --attributes-only don't copy the file data, just the attributes --backup[=CONTROL] make a backup of each existing destination file -b like --backup but does not accept an argument --copy-contents copy contents of special files when recursive -d same as --no-dereference --preserve=links --debug explain how a file is copied. Implies -v -f, --force if an existing destination file cannot be opened, remove it and try again (this option is ignored when the -n option is also used) -i, --interactive prompt before overwrite (overrides a previous -n option) -H follow command-line symbolic links in SOURCE -L, --dereference always follow symbolic links in SOURCE -P, --no-dereference never follow symbolic links in SOURCE --keep-directory-symlink follow existing symlinks to directories </pre>
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2. Basic Commands

Task	Command Name	Syntax	Example	Screenshots
To know today's date	date	date	date	
To print calendar	cal	cal	cal	
To print kernel version	uname	uname -r	uname -r	
To print default shell	echo	echo \$SHELL	echo \$SHELL	
To print current user	whoami	whoami	whoami	

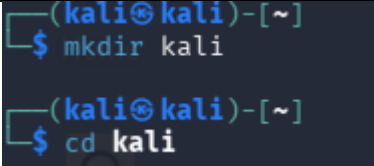
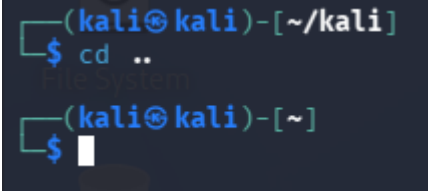
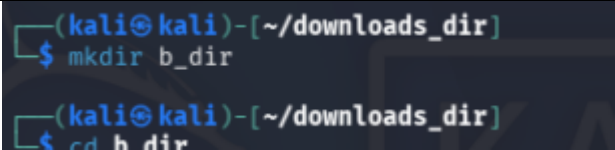
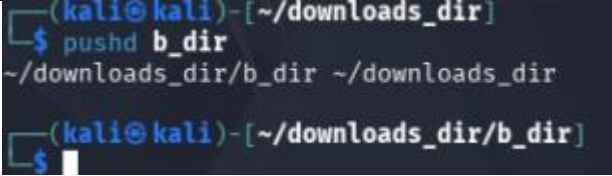
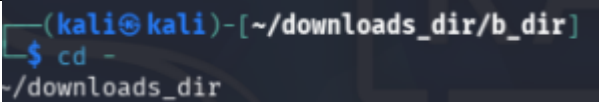
y logged in user				
To create shortcut for command	alias	alias name='command'	alias list='ls-i'	 <pre>(kali@kali)-[~] \$ alias list='ls -i' (kali@kali)-[~] \$ list 833156 aash 700667 Documents 833844 LinuxLab 700669 Pictures 700666 Public 700670 Videos 700663 Desktop 700664 Downloads 700668 Music 700671 Projects 700665 Templates</pre>
To delete shortcut	unalias	unalias name	unalias list	 <pre>(kali@kali)-[~] \$ unalias list (kali@kali)-[~] \$ list Command 'list' not found, did you mean: command 'gist' from deb yorick command 'hist' from deb loki command 'last' from deb wtmpdb command 'mlist' from deb mblaze command 'lift' from deb lift command 'flist' from deb nmh command 'dist' from deb nmh Try: sudo apt install <deb name></pre>
To change the timestamp of the file	touch	touch file	touch a.txt	 <pre>(kali@kali)-[~] \$ touch a.txt</pre>
To clear the screen	clear	clear	clear	 <pre>(kali@kali)-[~] \$ clear</pre>

To create empty files	touch	touch filename	touch b.txt	<pre>(kali@kali)-[~] \$ touch b.txt</pre>
To know disk usage	du	du -sh	du -sh	<pre>(kali@kali)-[~] \$ du -sh 3.0G .</pre>
To know free space in the system	df	df -sh	df -sh	<pre>(kali@kali)-[/] \$ df -h Filesystem Size Used Avail Use% Mounted on udev 888M 0 888M 0% /dev tmpfs 197M 984K 196M 1% /run /dev/sda1 79G 17G 58G 23% / tmpfs 985M 4.0K 985M 1% /dev/shm none 1.0M 0 1.0M 0% /run/credentials/systemd-journal ce tmpfs 985M 3.9M 981M 1% /tmp none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.serv tmpfs 197M 116K 197M 1% /run/user/1000</pre>
To know about the Linux release	cat	cat /etc/os-release	cat /etc/os-release	<pre>(kali@kali)-[~] \$ cat /etc/os-release PRETTY_NAME="Kali GNU/Linux Rolling" NAME="Kali GNU/Linux" VERSION_ID="2026.2" VERSION="2026.2" VERSION_CODENAME=kali-rolling ID=kali ID_LIKE=debian HOME_URL="https://www.kali.org/" SUPPORT_URL="https://forums.kali.org/" BUG_REPORT_URL="https://bugs.kali.org/" ANSI_COLOR="1;31"</pre>

To show login history	last	last	last	<pre> (kali@kali)-[~] \$ last kali tty7 :0 Thu Aug 20 11:42 - still logged in lightdm tty7 :0 Thu Aug 20 11:41 - 11:42 (00:00) kali tty7 :0 Tue Aug 18 00:34 - still logged in lightdm tty7 :0 Tue Aug 18 00:33 - 00:34 (00:00) kali tty7 :0 Thu Aug 13 18:38 - still logged in lightdm tty7 :0 Thu Aug 13 18:37 - 18:38 (00:00) kali tty7 :0 Tue Aug 11 00:46 - still logged in lightdm tty7 :0 Tue Aug 11 00:45 - 00:46 (00:00) kali tty7 :0 Mon Aug 10 19:12 - 20:17 (01:04) lightdm tty7 :0 Mon Aug 10 19:12 - 19:12 (00:00) kali tty7 :0 Mon Aug 10 08:25 - 09:07 (00:42) lightdm tty7 :0 Mon Aug 10 08:24 - 08:25 (00:00) kali tty7 :0 Thu Jul 30 12:34 - still logged in lightdm tty7 :0 Thu Jul 30 12:34 - 12:34 (00:00) postgres Thu Jun 16 10:29 - 10:29 (00:00) wtmptdb begins Tue Jun 16 10:29:31 2026 </pre>
To check the current user privileges	id	id	id	<pre> (kali@kali)-[~] \$ id uid=1000(kali) gid=1000(kali) groups=1000(kali), (users),101(netdev),102(scanner),118(wireshark) </pre>
To find about the Linux release	lsb_release	lsb_release -a	lsb_release -a	<pre> (kali@kali)-[~] \$ lsb_release -a No LSB modules are available. Distributor ID: Kali Description: Kali GNU/Linux Rolling Release: 2026.2 Codename: kali-rolling </pre>

To check system uptime	uptime	uptime	uptime	<pre>(kali㉿kali)-[~] \$ uptime 12:59:17 up 1:17, 1 user, load average: 0.23, 0.19, 0.17</pre>
To shutdown	shutdown	Sudo shutdown now	Sudo shutdown now	<pre>(kali㉿kali)-[~] \$ sudo shutdown now</pre>
To restart	reboot	sudo reboot	sudo reboot	<pre>(kali㉿kali)-[~] \$ <u>sudo</u> reboot</pre>

3. Navigation

Task	Command	Syntax	Screenshots
To navigate home directory	cd	cd	 <pre>(kali㉿kali)-[~] \$ mkdir kali (kali㉿kali)-[~] \$ cd kali</pre>
To navigate to the parent directory	cd ..	cd ..	 <pre>(kali㉿kali)-[~/kali] \$ cd .. (kali㉿kali)-[~] \$</pre>
To navigate to the child directory	cd	cd foldername	 <pre>(kali㉿kali)-[~/downloads_dir] \$ mkdir b_dir (kali㉿kali)-[~/downloads_dir] \$ cd b_dir</pre>
Alternate command to cd	pushd	Pushd directory	 <pre>(kali㉿kali)-[~/downloads_dir] \$ pushd b_dir ~/downloads_dir/b_dir ~/downloads_dir (kali㉿kali)-[~/downloads_dir/b_dir] \$</pre>
To go back to the previous directory	cd -	cd -	 <pre>(kali㉿kali)-[~/downloads_dir/b_dir] \$ cd - ~/downloads_dir</pre>

To go to the root directory	cd /	cd /	<pre>(kali㉿kali)-[~/downloads_dir] \$ cd / (kali㉿kali)-[/] \$</pre>
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4. File System

Task	Syntax	Command
How to identify the file system	df -T	df -T

5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

Use globbing patterns wherever possible, rather than specifying filenames individually.

- List all files ending in .html.

```
(kali㉿kali)-[~]
$ ls *.html
admin.html  backup1.html  index.html  login.html
```

- List all files other than .html files using extended globbing.

```
(kali㉿kali)-[~]
$ ls !(*.html)
zsh: unknown file attribute: h
```

- c. List filenames containing at least one numeric digit.

```
(kali㉿kali)-[~]  
$ ls *[0-9]  
ls: cannot access '*[0-9]': No such file or directory
```

- d. List files whose names begin with user followed by a single digit.

```
(kali㉿kali)-[~]  
$ ls user?.*  
user1.txt user2.log
```

- e. List files ending in either .txt or .log using a single extended glob pattern.

```
(kali㉿kali)-[~]  
$ ls *.@(txt|log)  
ls: cannot access '*.@(txt|log)': No such file or directory
```

- f. List files that do not end in .txt or .log.

```
(kali㉿kali)-[~]  
$ ls !(*.txt|*.log)  
ls: cannot access '!(*.txt|*.log)': No such file or directory
```

- g. List files beginning with either access or error using extended globbing.

```
(kali㉿kali)-[~]  
$ ls !(*.txt|*.log)  
ls: cannot access '!(*.txt|*.log)': No such file or directory
```

h. List .html files whose filenames contain a numeric digit.

```
(kali@kali)-[~]  
$ ls *[0-9]*.html  
backup1.html
```

6. Extended Globbing

a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

Write an extended glob pattern that selects only access.log, error.log, and auth.log.

```
(kali@kali)-[~]  
$ ls @(access|error|auth).log  
ls: cannot access '@(access|error|auth).log': No such file or directory
```

Select all files except notes.txt.

```
(kali@kali)-[~]  
$ ls !(notes.txt)  
zsh: unknown sort specifier
```

Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.

```
(kali@kali)-[~]  
$ ls auth.log.@(1|2)  
ls: cannot access 'auth.log.@(1|2)': No such file or directory
```

b. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log

construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

```
(kali㉿kali)-[~]  
$ ls malware*[0-9]*.log  
malware1.log  malware2.log
```

c. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log

use extended globbing to select .log files containing either failed_login or successful_login.

```
(kali㉿kali)-[~]  
$ ls @(failed_login|successful_login).log  
ls: cannot access '@(failed_login|successful_login).log': No such file or directory
```

Construct a pattern that lists all .log files except firewall.log.

```
(kali㉿kali)-[~]  
$ ls !("firewall.log")  
zsh: invalid mode specification
```

7. Filters

- a. Display the first 10 entries of login.log.

head -10 login.log

- b. Display the first 5 entries of access.log.

head -5 access.log

- c. Display the first 15 security events from security.log.

head -15 security.log

- d. Display the first 20 events recorded in incident.log.

head -20 incident.log

- e. Display only the first entry in login.log.

head -1 login.log

- f. Display the first 25 web requests recorded in access.log.

head -25 access.log

- g. Display the last 10 entries of login.log.

tail -10 login.log

- h. Display the last 5 entries of access.log.

tail -5 access.log

- i. Display the 20 most recent security events from security.log.

tail -20 security.log

- j. Display the 25 most recent events from incident.log.

tail -25 incident.log

- k. Display only the last entry in security.log.

tail -1 security.log

- l. Display the last 50 events recorded in incident.log.

tail -50 incident.log

- m. Display the contents of login.log starting from line 41 until the end of the file.

tail -n +41 login.log

- n. Display the contents of security.log starting from line 101 until the end of the file.

tail -n +101 security.log

- o. Find the total number of lines in login.log.

wc -l login.log

- p. Find the total number of lines in access.log.

wc -l access.log

q. Find the total number of security events in security.log.

```
wc -l security.log
```

r. Determine the total number of events recorded in incident.log.

```
wc -l incident.log
```

s. Count the number of words in login.log.

```
wc -w login.log
```

t. Count the number of bytes in security.log.

```
wc -c security.log
```

u. Display the line, word, and byte counts of access.log.

```
wc access.log
```

v. Find the line counts of all four .log files using a single wc command.

```
wc -l *.log
```

w. Display the first 20 entries of login.log and, from those entries, display only the last 5.

```
head -20 login.log | tail -5
```

x. Display the last 20 entries of login.log and, from those entries, display only the first 5.

```
tail -20 login.log | head -5
```

y. Display the first 30 entries of access.log and then display only the last 10 entries from that output.

```
head -30 access.log | tail -10
```

z. Display the last 30 entries of access.log and then display only the first 10 entries.

```
tail -30 access.log | head -10
```

aa. Display the first 50 events from security.log and then display the last 10 events from those 50.

```
head -50 security.log | tail -10
```

bb. Display the last 50 events from security.log and then display the first 10 events from those 50.

```
tail -50 security.log | head -10
```

cc. Display entries from line 11 to line 20 of login.log.

```
head -20 login.log | tail -10
```

dd. Display entries from line 21 to line 30 of login.log.

```
head -30 login.log | tail -10
```

ee. Display entries from line 11 to line 20 of access.log.

```
head -20 access.log | tail -10
```

ff. Display entries from line 31 to line 40 of access.log.

```
head -40 access.log | tail -10
```

gg. Display security events from line 21 to line 30 of security.log.

```
head -30 security.log | tail -10
```

hh. Display security events from line 51 to line 60 of security.log

head -60 security.log | tail -10.

ii. Display incident events from line 101 to line 110 of incident.log.

head -110 incident.log | tail -10

jj. Display incident events from line 201 to line 220 of incident.log.

head -220 incident.log | tail -20

kk. Display incident events from line 491 to line 500.

head -500 incident.log | tail -10

ll. Take the first 20 entries of login.log and count how many lines are displayed.

head -20 login.log | wc -l

mm. Take the last 15 entries of access.log and count how many lines are displayed.

tail -15 access.log | wc -l

nn. Take the first 50 entries of security.log and count the number of words in those entries.

head -50 security.log | wc -w

oo. Take the last 25 entries of security.log and count the number of lines.

tail -25 security.log | wc -l

pp. Extract lines 21–30 from login.log and count the number of extracted lines.

head -30 login.log | tail -10 | wc -l

qq. Extract lines 51–60 from access.log and count the number of extracted lines.

head -60 access.log | tail -10 | wc -l

rr. Extract events 101–150 from incident.log and count how many events are displayed.

head -150 incident.log | tail -50 | wc -l

ss. Display the first 100 events from incident.log, take the last 20 events from that output, and count the number of lines.

```
head -100 incident.log | tail -20 | wc -
```